

Kaushal Jain

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Portfolio: kaushaljain.com

GitHub: github.com/KaushalJainAI

Education

Punjab Engineering College, Chandigarh

2022–2026

B.Tech, Computer Science and Engineering (Data Science Specialization)

CGPA: **7.68**

Skills & Knowledge

- **CS & System Design:** DSA, OOP, OS, CN, DBMS, REST APIs, WebSockets, async workers, caching, auth, rate limiting
- **Languages & Tools:** Python · C++ · C · TypeScript · JavaScript · SQL · Git · Docker
- **Frameworks & Data:** Django REST · React · PostgreSQL · Redis · EDA · feature engineering · SQL query writing
- **Professional Skills:** Communication, technical documentation, debugging, AI-assisted development, and daily use of coding agents for faster prototyping and code review

Experience

Software Engineering Intern — Statcon Electronics

Jan 2025 – Jun 2025

- Built a Li-ion battery predictive-maintenance workflow using the NASA Battery Dataset; engineered time-series features and evaluated ML/DL models for RUL prediction.
- Developed real-time dashboards for 48V SMPS Power Plant monitoring and field-system analysis.
- Built a Django REST firmware-security platform with flash-count tracking, OTP-based access control, usage-based pricing, and automated billing workflows.

Projects

AIAAS — Agentic AI Workflow Automation Platform

2025–2026

aiaas.kaushaljain.com | [Backend](#) | [Frontend](#)

- Building an agentic workflow platform with a Django REST backend and React + TypeScript visual editor for creating, validating, saving, and executing node-based AI automations.
- Designed a compiler-style backend: converts ReactFlow workflow JSON into an executable graph using node indexing, DAG validation, reachability checks, config validation, expression resolution, and subworkflow support.
- Implemented a modular execution architecture with node handlers, shared workflow state, execution logs, heartbeat tracking, pause/resume controls, and WebSocket updates for long-running workflows.
- Added early support for multi-provider LLM routing, MCP tool access, encrypted credential storage, user-scoped credentials, human approval gates, and reusable skills/templates.
- Learned and applied system-design concepts through the project: separating compiler vs executor responsibilities, designing async execution flows, securing agent tool calls, handling WebSocket lifecycle issues, and avoiding credential leakage.

NGU Spices — Production E-Commerce Platform

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- Built and deployed a full-stack e-commerce platform for a spices brand with a Django REST backend, React storefront, and separate admin panel for products, orders, coupons, policies, and dashboard stats.
- Implemented catalog, combo offers, cart, checkout, order tracking, reviews, favorites, support chat, JWT authentication, password reset, and role-based admin access.
- Added AI-assisted product search using LLM-generated regional and phonetic spice synonyms with fuzzy matching, improving discovery for names like haldi, manjal, and other local-language variants.
- Optimized product and category APIs with Redis caching, namespaced cache keys, and signal-based invalidation for price, stock, category, and combo updates.
- Deployed with Docker, Nginx, AWS EC2, RDS/PostgreSQL, S3 media storage, and production environment configuration.

Research

- **Voice Disorder Classification — FEMH Dataset:** Built an ensemble model combining spectrogram-based CNN features with structured metadata for multi-class voice disorder classification.
- **Autism Detection using fMRI:** Developed an ASD detection pipeline using LSTMs and autoencoders for dimensionality reduction of high-dimensional neuroimaging sequences.
- **Ink Detection — Vesuvius Challenge:** Trained CNN-based segmentation models to detect hidden ink traces in low-contrast, multi-layer papyrus scans.

Achievements & Activities

- Solved 500+ problems on LeetCode across data structures, algorithms, and dynamic programming.
- PEC Music Club: 1st place in Battle of Bands (PEC Fest), 2nd place at Palaksha, and 3rd place at Amity Mohali.
- Active pianist, music composer, and sound engineer.